

PhD position in Computational Biocatalysis and Enzyme Design

The Computational Biochemistry and Biocatalysis (CompBioCat) research group at the **Institute of Computational Chemistry and Catalysis** (IQCC, <http://iqcc.udg.edu>), **University of Girona** (Spain), is seeking a highly motivated **PhD candidate** to join the ERC Consolidator Grant project **BioReInt** "Controlling Reactive Intermediates in Biocatalysis through Computational Design Tools", funded by the European Research Council.

Research Project. *BioReInt* aims to develop next-generation computational enzyme engineering approaches to exploit highly reactive intermediates in enzymatic catalysis. By combining quantum chemistry, molecular dynamics simulations, machine learning, and protein and enzyme engineering concepts, the project seeks to establish new design principles for creating biocatalysts capable of performing challenging and non-natural chemical transformations.

The selected PhD candidate will join the *CompBioCat* group, led by Dr. Marc Garcia-Borràs and Dr. Ferran Feixas, and will work in a dynamic, international and interdisciplinary environment at IQCC, interacting closely with experimental and computational collaborators worldwide and will receive training in state-of-the-art computational tools for biocatalysis and enzyme design.

Position details

- **Position:** Fully funded 4-year PhD position
- **Starting date:** Fall 2026, with 1 October 2026 as earliest possible start date
- **Candidate profile.** We are looking for candidates with a strong interest in computational chemistry, biochemistry, and biocatalysis, as well as enthusiasm for method development, scientific programming, and interdisciplinary research.
- **Requirements.** Applicants should have:
 - a. Completed 240 ECTS Bachelor degree + 60 ECTS Master Degree or 300 ECTS Bachelor degree or equivalent schemes in Chemistry, Computational Chemistry, Biotechnology, Biochemistry, Bioinformatics, Biophysics or a related discipline.
 - b. Background in computational chemistry
 - c. Basic understanding of organic chemistry and reaction mechanisms.
 - d. Good written and spoken communication skills in English.
 - e. Interest and previous experience in scientific programming, preferably in Python.
- **Additional skills and experience considered positively:**
 - a. Experience in quantum mechanical calculations (e.g. DFT)
 - b. Experience in molecular dynamics simulations
 - c. Familiarity with Linux/Unix environments and high-performance computing
 - d. Biocatalysis, enzyme engineering, or computational enzyme design.

How to apply. Applications should be addressed to Dr. Marc Garcia-Borràs (marc.garcia@udg.edu) and Dr. Ferran Feixas (ferran.feixas@udg.edu) via email by **September 15th 2026** including:

- a) Motivation letter highlighting the interest in the position.
- b) Curriculum vitae with emphasis on previous research experience.
- c) Name and contact address of at least one reference.

Selected references:

- [1] Soler, J.; Gergel, S.; Klaus, C.; Hammer, S. C.; Garcia-Borràs, M.; "Enzymatic Control over Reactive Intermediates Enables Direct Oxidation of Alkenes to Carbonyls by a P450 Iron-Oxo Species" *J. Am. Chem. Soc.* **2022**, 144, 35, 15954–15968
- [2] Calvó-Tusell, C.; Liu, Z.; Chen, K.; Arnold, F. H.; Garcia-Borràs, M.; "Reversing the Enantioselectivity of Enzymatic Carbene N-H Insertion Through Mechanism-Guided Protein Engineering" *Angew. Chem. Int. Ed.* **2023**, e202303879.
- [3] Feixas F.; Garcia-Borràs, M. (2025) "Multiscale Computational Modeling for the Discovery and Characterization of New Metalloenzyme-catalyzed Reactions" *Computational Bioinorganics: From Description to Prediction*, John Wiley & Sons <https://doi.org/10.1002/9781119415152.ch05>